

LAB TASK#12

NAME: MASHAAL ISHTIAQ
ROLL#: 22P-9326
SECTION:BCS-4C

CODE:

```
        #include <iostream>
#include <string>
using namespace std;

// Define the student structure
class Student
{

public:
   string rollNumber;
   string name;
   string fatherName;
   string department;
   string degree;
   Student *next;

   Student(string rn, string n, string fn, string dep, string deg)
       : rollNumber(rn), name(n), fatherName(fn), department(dep), degree(deg),
next(nullptr) {}
};

//__ hash table class for student management
class StudentHashTable
{
private:
   Student **table; // handling collisions using chaining
   int capacity;    // Size of the hash table

   int hash(string rollNumber)
   {
       return stoi(rollNumber.substr(0, 2)); // Extract the two-digit year and
convert from stringt->integer
   }

public:
   StudentHashTable(int cap) : capacity(cap)
   {
       table = new Student *[capacity]();
   }

   //____Insert
```

```
    void insert(Student *student)
    {
        int index = hash(student->rollNumber);
        student->next = table[index];
        table[index] = student;
    }

    //___Search

    Student *search(string rollNumber)
    {
        int index = hash(rollNumber);
        Student *current = table[index];
        while (current != nullptr)
        {
            if (current->rollNumber == rollNumber)
                return current;
            current = current->next;
        }
        return nullptr; // Return nullptr if no student found with the given roll
number
    }

    //___display

    void display()
    {
        for (int i = 0; i < capacity; ++i)
        {
            cout << "Year " << i << ": ";
            Student *current = table[i];
            while (current != nullptr)
            {
                cout << current->name << " (" << current->rollNumber << ") ";
                current = current->next;
            }
            cout << endl;
        }
    }

    //___deleting the dynamically allocated memory
    ~StudentHashTable()
    {
        for (int i = 0; i < capacity; ++i)
        {
```

```
            Student *current = table[i];
            while (current != nullptr)
            {
                Student *temp = current;
                current = current->next;
                delete temp;
            }
        }
        delete[] table;
    }
};

int main()
{
    StudentHashTable ht(100); // Create a hash table with 100 slots for years 00 ->
99

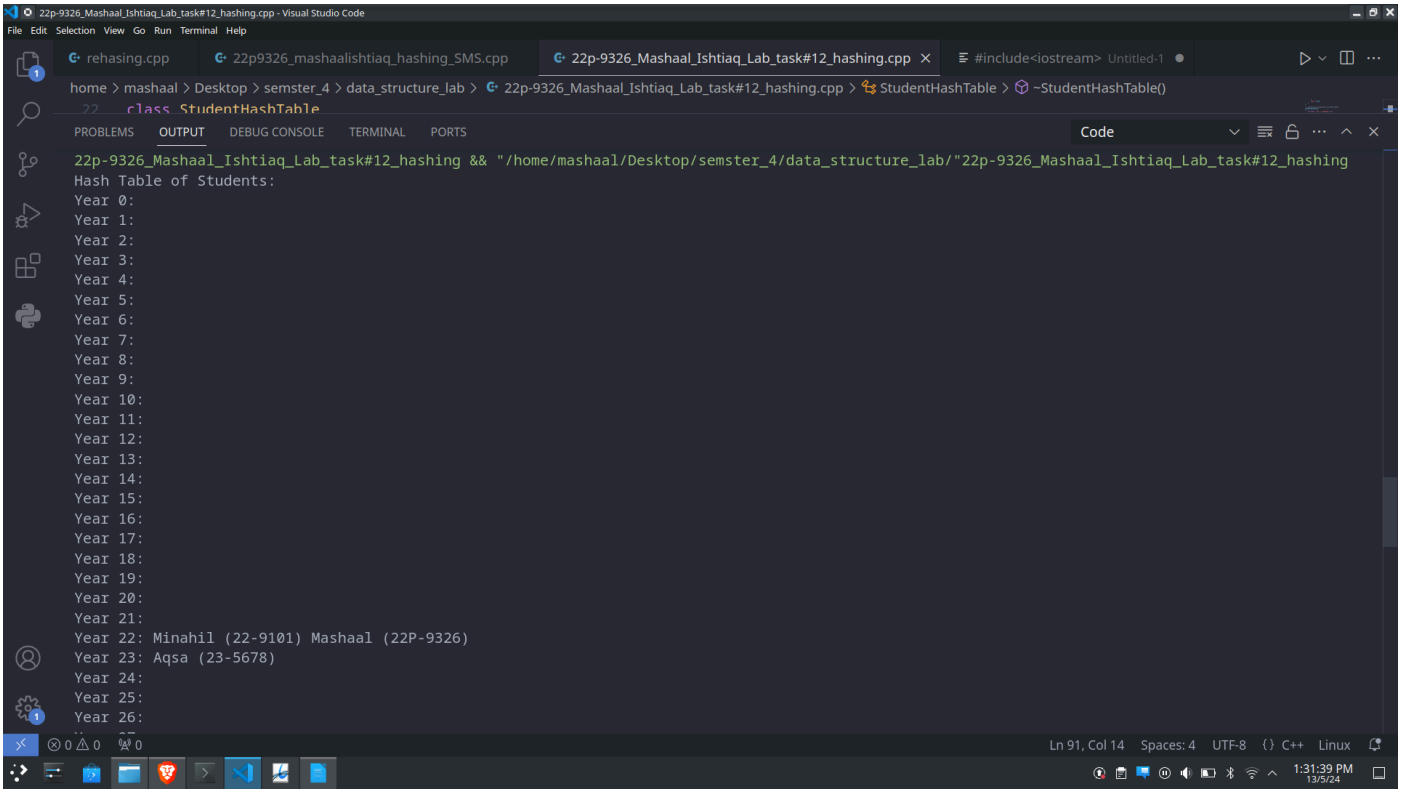
    //__ inserting some student records
    ht.insert(new Student("22P-9326", "Mashaal", " Ishtiaq", "Software Engeering ",
"BSE"));
    ht.insert(new Student("23-5678", "Aqsa", "yousaf", "Aritifical intelligence",
"BAI"));
    ht.insert(new Student("22-9101", "Minahil", "Fatima", "Mechanical Engineering",
"BSME"));

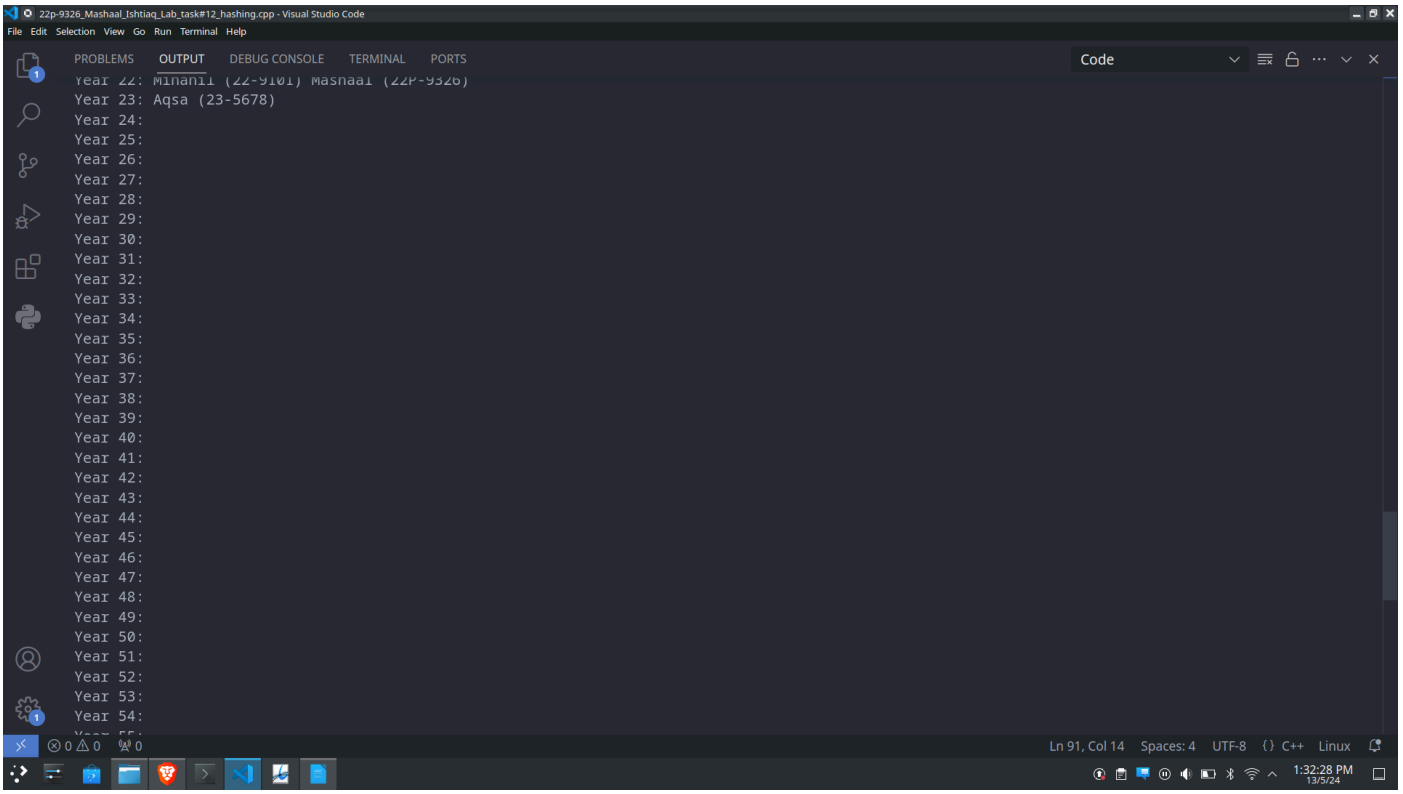
    //___ Display all students
    cout << "Hash Table of Students:\n";
    ht.display();

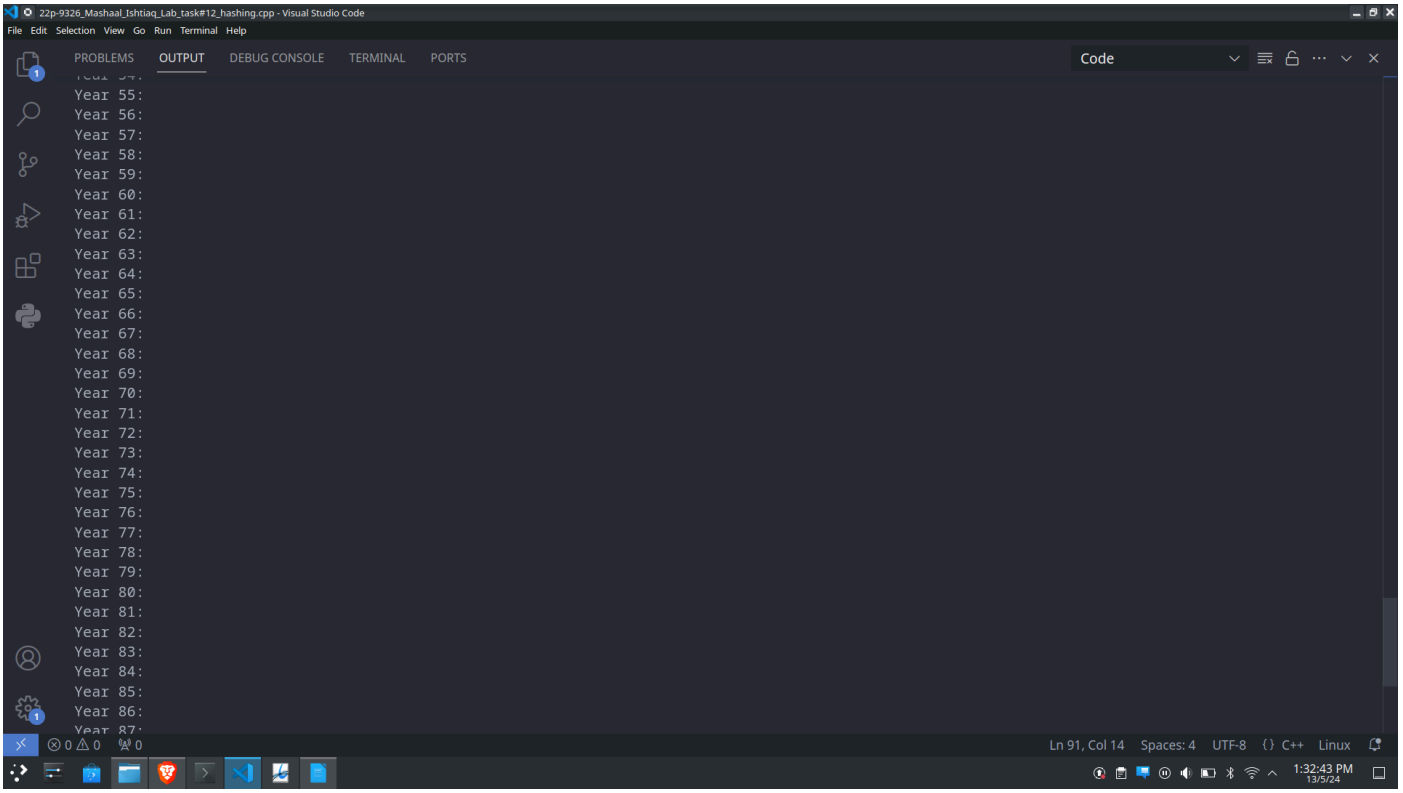
    //____ Searching a student

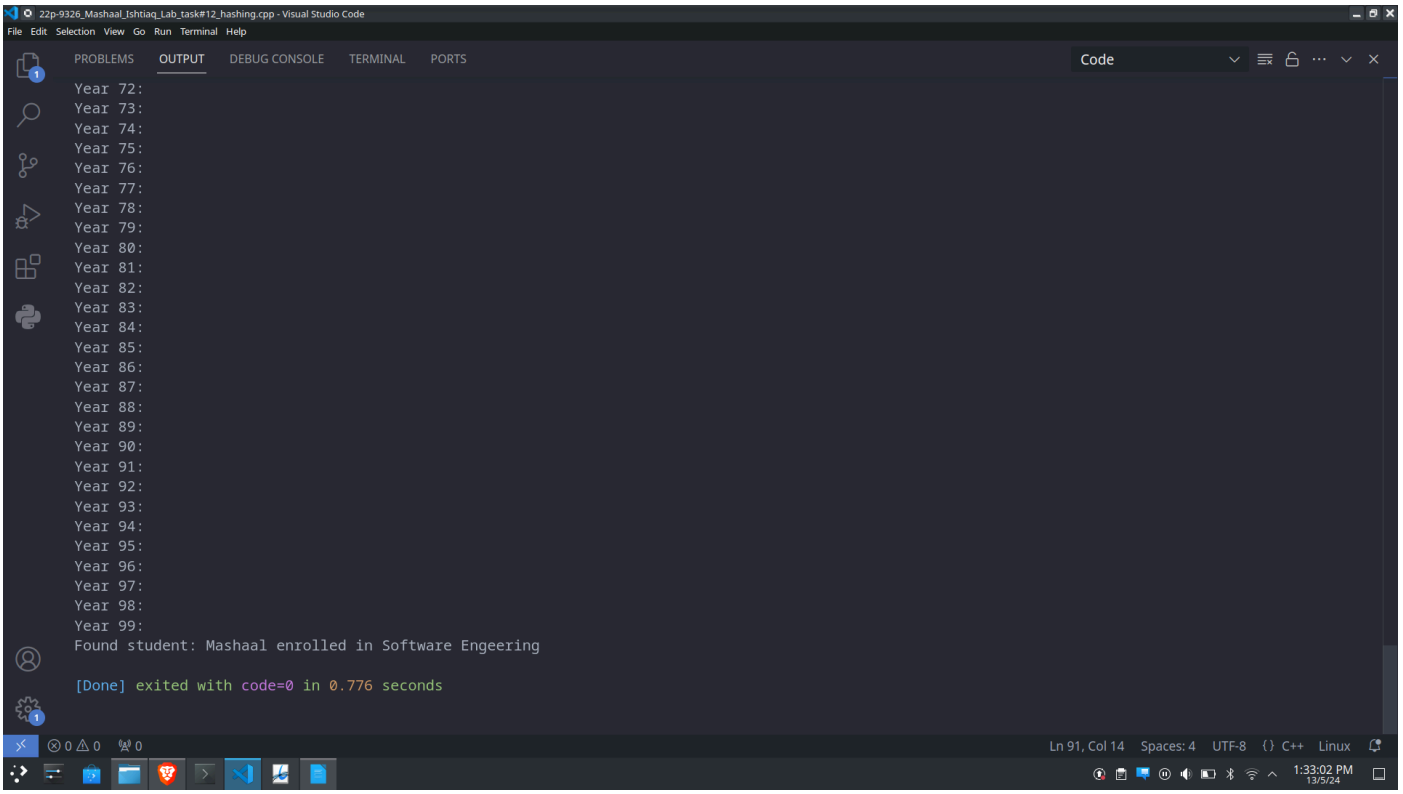
    string roll = "22P-9326";
    Student *foundStudent = ht.search(roll);
    if (foundStudent)
    {
        cout << "Found student: " << foundStudent->name << " enrolled in " <<
foundStudent->department << endl;
    }
    else
    {
        cout << "Student with roll number " << roll << " not found." << endl;
    }

    return 0;
}
```


22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing.cpp - Visual Studio Code
File Edit Selection View Go Run Terminal Help
rehasing.cpp
22p9326_mashaalishtiaq_hashing_SMS.cpp
22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing.cpp
#include<iostream> Untitled-1
home > mashaal > Desktop > semster_4 > data_structure_lab > 22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing.cpp > StudentHashTable > ~StudentHashTable()
class StudentHashTable
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing && "/home/mashaal/Desktop/semster_4/data_structure_lab/"22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing
Hash Table of Students:
Year 0:
Year 1:
Year 2:
Year 3:
Year 4:
Year 5:
Year 6:
Year 7:
Year 8:
Year 9:
Year 10:
Year 11:
Year 12:
Year 13:
Year 14:
Year 15:
Year 16:
Year 17:
Year 18:
Year 19:
Year 20:
Year 21:
Year 22: Minahil (22-9101) Mashaal (22P-9326)
Year 23: Aqsa (23-5678)
Year 24:
Year 25:
Year 26:
Ln 91, Col 14 Spaces: 4 UTF-8 C++ Linux
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22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing.cpp - Visual Studio Code
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
Year 22: Minahil (22-9101) Mashaal (22P-9326)
Year 23: Aqsa (23-5678)
Year 24:
Year 25:
Year 26:
Year 27:
Year 28:
Year 29:
Year 30:
Year 31:
Year 32:
Year 33:
Year 34:
Year 35:
Year 36:
Year 37:
Year 38:
Year 39:
Year 40:
Year 41:
Year 42:
Year 43:
Year 44:
Year 45:
Year 46:
Year 47:
Year 48:
Year 49:
Year 50:
Year 51:
Year 52:
Year 53:
Year 54:
Ln 91, Col 14 Spaces: 4 UTF-8 C++ Linux
1:32:28 PM
13/5/24


22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing.cpp - Visual Studio Code
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
Year 55:
Year 56:
Year 57:
Year 58:
Year 59:
Year 60:
Year 61:
Year 62:
Year 63:
Year 64:
Year 65:
Year 66:
Year 67:
Year 68:
Year 69:
Year 70:
Year 71:
Year 72:
Year 73:
Year 74:
Year 75:
Year 76:
Year 77:
Year 78:
Year 79:
Year 80:
Year 81:
Year 82:
Year 83:
Year 84:
Year 85:
Year 86:
Ln 91, Col 14 Spaces: 4 UTF-8 C++ Linux
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22p-9326_Mashaal_Ishtiaq_Lab_task#12_hashing.cpp - Visual Studio Code
File Edit Selection View Go Run Terminal Help
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Code
Year 72:
Year 73:
Year 74:
Year 75:
Year 76:
Year 77:
Year 78:
Year 79:
Year 80:
Year 81:
Year 82:
Year 83:
Year 84:
Year 85:
Year 86:
Year 87:
Year 88:
Year 89:
Year 90:
Year 91:
Year 92:
Year 93:
Year 94:
Year 95:
Year 96:
Year 97:
Year 98:
Year 99:
Found student: Mashaal enrolled in Software Engeering
[Done] exited with code=0 in 0.776 seconds
Ln 91, Col 14 Spaces: 4 UTF-8 C++ Linux
1:33:02 PM
13/5/24